NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur
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Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 3 - Week 3 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

If the current block reward for Bitcoin is 3.125 BTC, and the network remains calibrated so that a new block is found every 10 minutes on average, how much Bitcoin will be mined in a total 24-hour period?

a) 312.5 BTC
b) 450 BTC
c) 252 BTC
d) 900 BTC

Answer: (b)

Detailed solution:
The block reward is **3.125 BTC**.
There are **144 blocks** mined in a 24-hour period (24 hours × 60 minutes / 10 minutes per block).
In one day, the total Bitcoin mined would be 144 × 3.125 = 450 BTC.

QUESTION 2

What is a fundamental difference between a **permissionless** (public) blockchain and a **permissioned** blockchain?

a) Permissioned blockchains do not use a distributed ledger.
b) Permissionless blockchains require participants to prove their identity to a central authority before they can join the network.
c) Permissioned blockchains utilize a gatekeeping mechanism that restricts the ability to validate transactions and update the ledger to a set of authorized nodes.
d) Permissioned blockchains allow any user to participate in the consensus process without prior authorization.

Answer: (c)

Detailed solution:
Permissioned blockchains use an **access control layer** to restrict ledger validation to a set of pre-approved, authorized nodes, unlike open permissionless networks.

QUESTION 3

Which of the following combinations is correctly used to compute Bitcoin's current block hash?

a) Previous block's hash, Merkle root, block reward, nonce, timestamp, and block size
b) Previous block's hash, timestamp, nonce, Merkle root, difficulty bits, and block version
c) Block creator's public key, Merkle root, timestamp, block reward, nonce, and difficulty level
d) Previous block's hash, nonce, Merkle root, height, timestamp, and difficulty bits

Answer: (b)

Detailed solution:
Please refer to the Week 3 Lecture 13.

QUESTION 4

Which of the following difficulty targets would make it most difficult for miners to find a valid block?

a) 0000000000000000000000000000000000000000000000000000056789abcdef12
b) 0000000000000000000000000000000000000000000000000002dfe34a1b2c3d55
c) 0000000000000000000000000000000000000000000000000000afc123987ab
d) 00000000000000000000000000000000000000000000000000000000000af467

Answer: (d)

Detailed solution:
The difficulty target with **59 leading zeros** (option **d**) is the most difficult for miners to find a valid block. Therefore, 00000000000000000000000000000000000000000000000000000000000af467 is the correct choice.

QUESTION 5

In the Bitcoin, block identifier refers to

a) SHA1 (128 bits) of the future block header
b) Double SHA256 of the current block header
c) Double SHA256 of the difficulty bits only
d) Triple SHA256 of the future block header

Answer: (b)

Detailed solution:
The Bitcoin block identifier (block hash) contains **Double SHA256** on the current block header. This means performing SHA256 twice. Please refer to the Week 3 Lecture 13.

QUESTION 6

If a Bitcoin block contains 64 transactions, how many hashing steps (Merkle path length) are required to verify the inclusion of a specific transaction without downloading the entire block?

a) 64
b) 32
c) 6
d) 12

Answer: (c)

Detailed solution:
The number of hashes required for a Merkle proof is proportional to the height of the tree, calculated as $log_2(n)$. For 64 transactions: $log_2(64) = 6$.

QUESTION 7

In Bitcoin Script, the stack is shown from bottom to top, left to right.
The current stack is: `Bottom → [ x1, x2, x3 ] ← Top`
Which opcode will transform the stack into: `Bottom → [ x1, x3 ] ← Top`

a) OP_DROP
b) OP_NIP
c) OP_2DROP
d) OP_DEPTH

Answer: (b)

Detailed solution:
OP_DROP removes the top item (x3). **OP_NIP** removes the *second-to-top* item (x2), leaving the top item intact. **OP_2DROP** removes the top two items. **OP_DEPTH** pushes the stack size onto the stack.

QUESTION 8

If a Merkle tree is constructed from 16 transactions, what is the total number of hashes (nodes) in the entire tree, including the transaction hashes (leaves) and the Merkle root?

a) 32
b) 16
c) 31
d) 15

Answer: (c)

Detailed solution:
A Merkle tree is a binary tree structure. To find the total nodes, you sum the number of hashes at each level:

1. Level 0 (Leaves): 16 hashes (one for each transaction).
2. Level 1: The 16 leaves form 8 pairs → 8 hashes.
3. Level 2: The 8 hashes form 4 pairs → 4 hashes.
4. Level 3: The 4 hashes form 2 pairs → 2 hashes.
5. Level 4 (Root): The 2 hashes form 1 Root hash.

Total: 16 + 8 + 4 + 2 + 1 = 31 hashes.

(Note: In a perfect binary tree, the total number of nodes is always $2n - 1$, where n is the number of leaves. Here, $2(16) - 1 = 31$.)

QUESTION 9

The "Nonce" field in the Bitcoin block header is 32 bits long. What is the maximum value a miner can test in the nonce field before needing to change other header fields (like the timestamp or extra nonce)?

a) 2^{64} (approx 1.84 x 10^{19})
b) 2^{32} (approx 4 billion)
c) 2^{256} (approx 1.15 x 10^{77})
d) 1,000,000

Answer: (b)

Detailed solution:
The nonce is a value miners iterate through6. Since it is a 32-bit field, the maximum number of values is 2^{32} (approximately 4.29 billion). Once a miner exhausts all these values without finding a valid block, they must alter other parts of the header (like the timestamp or the coinbase transaction extra nonce) to generate new hashes.

QUESTION 10

What happens if the number of transactions in a Merkle tree is odd?

a) The tree cannot be built
b) Dummy (duplicate) hashes are added to adjust
c) Transactions are left out of the block
d) The Merkle root is ignored

Answer: (b)

Detailed solution:
If the number of transactions is odd, pairs cannot be formed for the suitable structure of the Merkle tree. To fix this, dummy (duplicate) hashes are added to make the number of transactions even, allowing the tree to be constructed properly. Please refer to the Week 3 Lecture 14.